INCH-POUND
MIL-I-45932/38
2 MAY 1994
SUPERSEDING
MIL-I-45932/3A
1 MARCH 1989

MILITARY SPECIFICATION SHEET

INSERT, SCREW THREAD - THIN WALL, LOCKED IN, OVERSIZE REPLACER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-45932

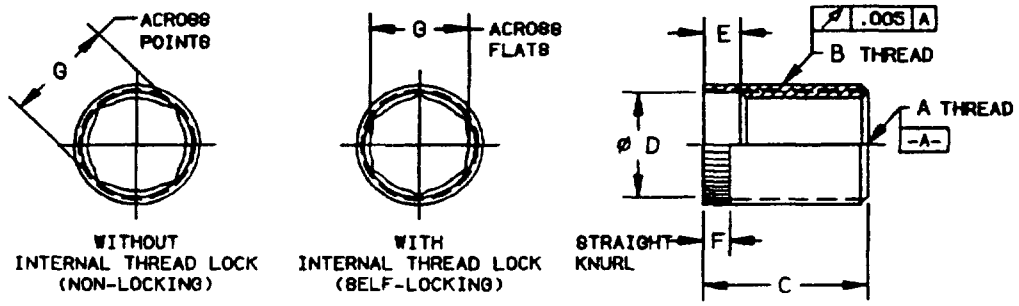


TABLE I. Dash numbers and characteristics

DASH NUMBERS (REQT 7)				A INT THD CLASS 3B (REQT 4)	B EXT THREAD ALTERED MINOR DIA		C	ØD	E	F	G	MIN SHEAR ENGAGEMENT AREA SQ IN.
17-4PH CRES 1/	ALLOY STEEL 1/	A286 CRES 1/ SILVER PLATED	A286 CRES 1/ SOLID FILM LUBE		THREAD SIZE	MAX MINOR DIA	±.010	+.008 -.002	+.015 -.000	(REF)	(REF)	(NOTE 2)
3 L	3 AL	3 CL	3 DL	0.1120-40 UNC	0.1900-32 UNF	.1620	.190	.142	060	.045	.092	.0439
4	4 A	4 C	4 D								.100	
5 L	5 AL	5 CL	5 DL	0 1380-32 UNC	0.2160-28 UNF	.1758	.210	142	080	.055	.113	.0542
6	6 A	6 C	6 D								.120	
7 L	7 AL	7 CL	7 DL	0.1640-32 UNC	0.2500-28 UNF	.2098	.250	.169	.080	.060	.138	.0871
8	8 A	8 C	8 D								.150	
9 L	9 AL	9 CL	9 DL	0.1900-32 UNF	0.2812-28 UNS	.2410	.290	214	080	075	.157	1147
10	10 A	10 C	10 D								.180	
11 L	11 AL	11 CL	11 DL	0 1900-24 UNC							.157	
12	12 A	12 C	12 D								.180	
13 L	13 AL	13 CL	13 DL	0.2500-28 UNF	0.3438-24 UNS	.2976	.380	264	.095	.075	.210	.2153
14	14 A	14 C	14 D								.240	
15 L	15 AL	15 CL	15 DL	0 2500-20 UNC							.210	
16	16 A	16 C	16 D								240	
17 L	17 AL	17 CL	17 DL	0.3125-24 UNF	0.4219-20 UNS	3651	470	336	.110	075	.266	.3591
18	18 A	18 C	18 D								.310	
19 L	19 AL	19 CL	19 DL	0.3125-18 UNC							.266	
20	20 A	20 C	20 D								.310	
21 L	21 AL	21 CL	21 DL	0.3750-24 UNF	0.4844-20 UNS	.4276	.560	.393	.110	.105	.322	.4938
22	22 A	22 C	22 D								.370	
23 L	23 AL	23 CL	23 DL	0.3750-16 UNC							.322	
24	24 A	24 C	24 D								.370	
25 L	25 AL	25 CL	25 DL	0.4375-20 UNF	0.5626-18 UNF	.4993	.660	.466	.135	.105	.377	.6714
26	26 A	26 C	26 D								.430	
27 L	27 AL	27 CL	27 DL	0.4375-14 UNC							.377	
28	28 A	28 C	28 D								.430	
29 L	29 AL	29 CL	29 DL	0.5000-20 UNF	0.6250-18 UNF	.5618	.750	.528	.135	.105	.439	.8717
30	30 A	30 C	30 D								.490	
31 L	31 AL	31 CL	31 DL	0.5000-13 UNC							.439	
32	32 A	32 C	32 D								490	

1/ "L" Suffix shown indicates self-locking insert.

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REQUIREMENTS:

1. Material:
Steel, alloy, grade 4130 (UNS G41300) per AMS6370 or grade 8740 (UNS G87400) per AMS6322
Steel, corrosion-resistant, type 17-4 PH (UNS S17400) per AMS5643.
Steel, corrosion-resistant, type A286 (UNS S66286) per AMS5731 or AMS5734.

2. Protective coating or treatment:
Steel, alloy, shall be cadmium plated per QQ-P-416 type III class 3, plus solid film lubricant coating per MIL-L-46010, type I
Steel, corrosion-resistant, type 17-4 PH, shall be solid film lubricant coated per MIL-L-46010, type I.
Steel, corrosion-resistant, type A286,
Dash C & CL shall be silver plated per AMS2411 grade B, .0002 thick min
Dash D & DL shall be solid film lubricant coated per MIL-L-46010, type I

3. Surface roughness:
Machined surfaces shall be 125 microinches in accordance with ANSI/ASME B46.1 except knurling.

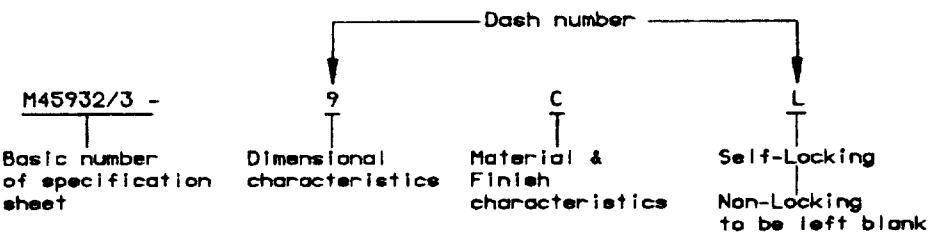
4 Threads:
Threads shall be in accordance with MIL-S-7742 except as noted in table I and shall accept external MIL-S-8879 threads. All coarse internal threads have an increased minor diameter. Threads are prior to the addition of solid film lubricant.

5 Hardness:
Alloy steel, 25-34 HRC
Corrosion-resistant steel, 17-4 PH, 35-42 HRC
Corrosion-resistant steel, A286, 32-40 HRC

6. Internal thread locking feature.
The centerline of the internal thread locking feature shall be approximately mid-length of internal thread.

7. Part Identifying Number (PIN).
Consists of the letter M, the basic number of the specification sheet, and a dash number taken from table I

Example of PIN



M45932/3-9CL Insert, Screw Thread - Thin Wall, Locked In, Oversize Rplacer, 0.1900-32 UNF-3B Internal Thread, A286 Corrosion Resistant Steel, Silver Plated, Self-Locking

M45932/3-10D Insert, Screw Thread - Thin Wall, Locked In, Oversize Replacer, 0.1900-32 UNF-3B Internal Thread, A286 Corrosion Resistant Steel, Solid Film Lubricant Coated, Non - Locking.

NOTES:

1. Dimensions.
All dimensions are in inches, to be met after plating and before the addition of solid film lubricant (see requirement 2 herein).

2 Shear engagement area:
Shear engagement area is the assembled dimensional value for the overall engaged area of mating thread members. It does not represent a dimension of either of the members in an unassembled condition.

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HOLE PREPARATION, INSTALLATION & REMOVAL REQUIREMENTS

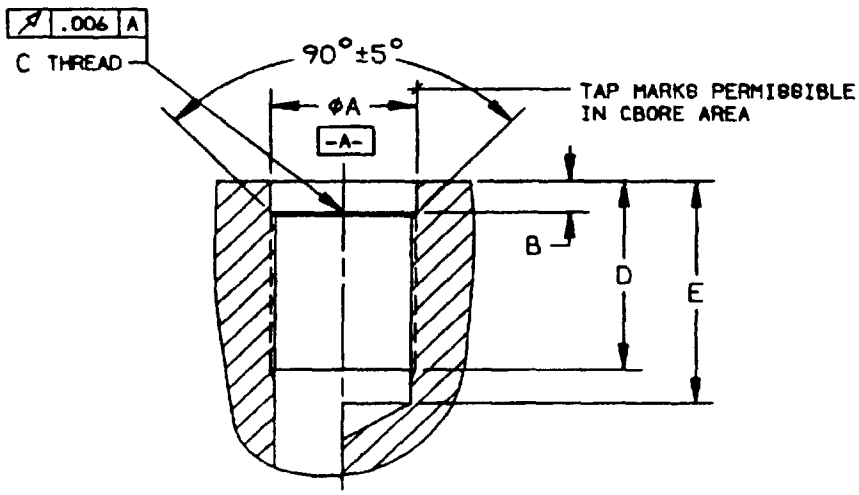


TABLE II. Installation & removal criteria

NOMINAL EXTERNAL THREAD SIZE OF INSERT (REF)	INSERT DASH NUMBER M45932/3 (REF)	ØA CBORE + .004 -.001	B CBORE DEPTH (NOTE 4a) ±.005	C THREAD MIL-S-8879		D MINIMUM FULL THREAD DEPTH	E MINIMUM DRILL DEPTH BLIND HOLE	INSERT REMOVAL DRILL SIZE (NOTE 5)
				CLASS-3B EXCEPT MINOR Ø	CONTROLLED MINOR Ø			
0.1900-32	3 4	.187	.065	0.1900-32 UNJF	.165-.170	.220	.298	#17
0.2160-28	5 6	216	.065	0 2160-28 UNJF	181-.186	.240	.329	#5
0.2500-28	7 8	.250	065	0.2500-28 UNJF	.217-.222	280	.369	15/64
0.2812-28	9 10 11 12	.281	.082	0.2812-28 UNJS	.241-.246	.325	.414	17/64
0.3438-24	13 14 15 16	.343	.082	0.3438-24 UNJS	.301-.306	415	.519	21/64
0.4219-20	17 18 19 20	.422	.082	0.4219-20 UNJS	.367-.372	.505	.630	13/32
0.4844-20	21 22 23 24	.484	.113	0.4844-20 UNJS	.436-.441	595	.720	29/64
0.5625-18	25 26 27 28	.562	.113	0.5625-18 UNJF	.514-.519	.695	.834	35/64
0.6250-18	29 30 31 32	.625	.113	0.6250-18 UNJF	577- 582	785	.924	39/64

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NOTES:

1. Axis of hole shall be normal to entry surface or provide spot face when required.

2. Machined surfaces shall be 125 microinches in accordance with ANSI/ASME B46.1

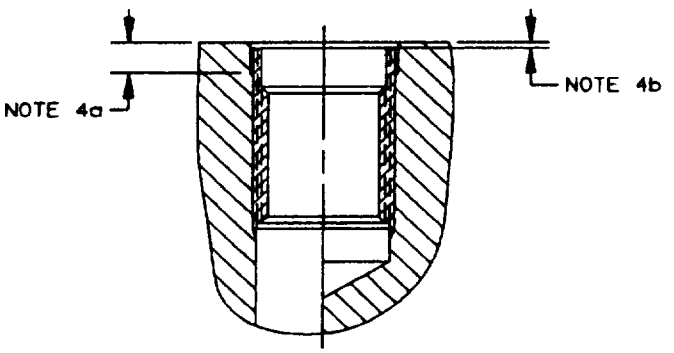
3. All dimensions are in inches

4. Install insert:
   (a) These inserts are primarily designed for use in aluminum, magnesium and other non-ferrous materials that do not exceed 187 HB (3000 kg load and 10 mm ball). Use in corrosion-resistant steels, titanium and hardened ferrous materials will require broaching serrations in counterbore to accept the insert knurls during swaging operation. Installation in steel will also require counterbore depth "B" in table II to be increased by .015 inches

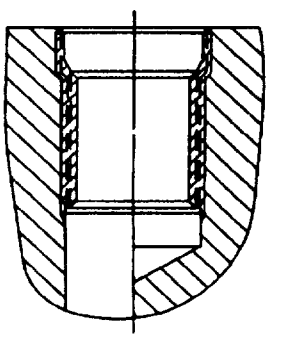
   (b) Install inserts -3 thru -8 into hole until the top of insert is .010- 020 below boss surface and -9 thru -32 inserts .015- 025 below boss surface

   (c) Place swage tool in insert and apply a downward force sufficient to effect full swageout and external lock setting.

INSERT INSTALLATION



VIEW
INSERT BEFORE SWAGING



VIEW
INSERT AFTER SWAGING

5. Replacement of inserts are made with same size inserts as those removed. Using removal drill size shown in table II, drill to depth "B" + .025 then back-out insert using installation wrench or a square type screw extractor. Remove loose chips, re-inspect hole and then re-install per note 4.

Custodians:
Army - AR
Navy - AS
Air Force - 99

Review activities:
Army - AT, AV, ER, ME, MI
Navy - MC, OS, YD
DLA - IS
NSA - NS

Preparing activity:
Army - AR

(Project 5340-2197)

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